

中数错题本 ¹ 

A4 Spiral Notebook



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Feint Ruled



55
GSM



120
Pages

120

中数

Question

中数 — Trigonometric Symmetry properties 2.2

↑↑↑↑↑↑↑↑↑↑ (43连击) 25/05/2022 T ↑↑↑↑↑↑↑↑↑↑

Q1 Simplify $\sin(270^\circ - \alpha)$ Knowledge

$$\cos\left(\frac{\pi}{2} - \theta\right) = \sin(\theta)$$

$$\sin\left(\frac{\pi}{2} - \theta\right) = \cos(\theta)$$

$$\tan\left(\frac{\pi}{2} - \theta\right) = \frac{1}{\tan(\theta)}$$

↓↑↑↑↑↑↑↑↑↑ (43连击) 25/05/2022 T ↑

Q2. $\cos(\alpha) = 0.8572$ in Quadrant 1, find $\tan(180^\circ - \alpha)$, to 4 decimal Thinking
(Active.)

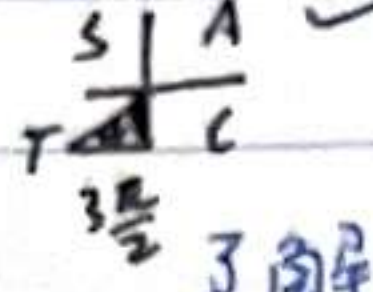
Over

Answers

错误思路: 将 $270^\circ - \alpha$ 变成 $180^\circ - (90^\circ - \alpha)$ 等复杂想法

正确思路: $\sin(270^\circ - \alpha) = \sin(\frac{3\pi}{2} - \alpha)$

用 $\sin(\frac{3\pi}{2} - \alpha) = -\cos(\alpha)$



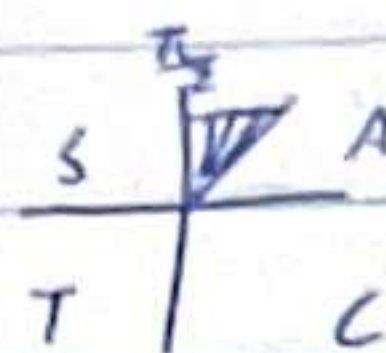
Q1 quadrant

$\cos(\frac{\pi}{2} - \theta) = \sin(\theta)$

$\sin(\frac{\pi}{2} - \theta) = \cos(\theta)$

$\tan(\frac{\pi}{2} - \theta) = \frac{1}{\tan(\theta)}$ 是基本

$\sin(\frac{3\pi}{2} - \alpha)$ 在 Quadrant 3, 此 quadrant, $\sin$ 是负数, 因此要在基本 $\cos(\alpha)$ 前加负号。



错误思路: 卡在第二步, 未能将其他三角公式与已有条件结合

正确思路:

① Convert $\tan(180^\circ - \alpha)$: $\tan(180^\circ - \alpha) = -\tan(\alpha)$

② Represent $\tan(\alpha)$: 找 $\sin(\alpha)$: $\sin^2(\alpha) + \cos^2(\alpha) = 1$

with $\cos(\alpha), \sin(\alpha)$

$\sin^2(\alpha) = 1 - 0.8572^2$

[2018年]

$\pi \leq \alpha < 2\pi$

(2018年)

$\sin(\alpha) = \pm \sqrt{1 - 0.8572^2}$

negative rejected in Q1

≈ 0.5150

③ Find $-\tan(\alpha)$: $\tan(\alpha) = \frac{0.5150}{0.8572}$

≈ 0.6008

$-\tan(\alpha) = -0.6008$

$\pi \leq \alpha < 2\pi$ 所以 $\sin(\alpha) = -\sqrt{1 - 0.8572^2}$ (2018年) ④

Questions

中数 - Trigonometric equations 2.3

① $2 \sin(2x + \pi) + \sqrt{3} = 0$ for $-\pi \leq x \leq \pi$ [标准问题]

② Solve $\cos^2(\theta) - \sin(\theta)\cos(\theta) = 0$ for $0 \leq \theta \leq 2\pi$ @question understanding.
* 错误: 忘记求 $\cos(\theta) = 0$

③ Solve $\sin^2(\alpha) = \cos^2(\alpha)$ $-\pi \leq \alpha \leq \pi$ [程序化]

④ Use CAS Solve $2 \sin(2x + \frac{\pi}{4}) = \sqrt{2}$ for $-\pi \leq x \leq \pi$

Answers.

○ ① 将 $\sin(2x+\pi)$ 隔离: $\sin(2x+\pi) = \frac{-\sqrt{3}}{2}$ $\frac{\pi}{3}$ $\frac{\pi}{c}$ Q3+Q4

② change the domain: $-\pi \leq x \leq \pi$ 乘 2

$-2\pi \leq 2x \leq 2\pi$ 加 π

$-\pi \leq 2x+\pi \leq 3\pi$

③ 计算 $2x+\pi$ 的值: $2x+\pi = \pi + \frac{\pi}{3}, 2\pi - \frac{\pi}{3}, -\frac{\pi}{3}, -\frac{2\pi}{3}, \frac{10\pi}{3}$ rejected
 $= \frac{4\pi}{3}, \frac{5\pi}{3}, -\frac{\pi}{3}, -\frac{2\pi}{3}$

④ 计算 x 的值: $x = (\frac{4\pi}{3} - \pi) \times \frac{1}{2}, \dots$
 $= \frac{\pi}{6}, \frac{\pi}{3}, -\frac{2\pi}{3}, -\frac{5\pi}{6}$

○ ① 简化 $\cos^2(\theta) - \sin(\theta)\cos(\theta) = 0$: $\cos(\theta)(\cos(\theta) - \sin(\theta)) = 0$

得出两个公式: $\cos(\theta) = 0$ 和 $\cos(\theta) = \sin(\theta)$
 求 x 的 $\leftarrow$ 设两边为 0 $\leftarrow$

② 算 $\cos(\theta) - \sin(\theta) = 0$ 的 θ 值: $\frac{\sin(\theta)}{\cos(\theta)} = 1$

$\tan(\theta) = 1$ $\frac{\pi}{4}$ $\frac{\pi}{c}$ Q1, Q3

$0 \leq \theta \leq 2\pi$ $\theta = \frac{\pi}{4}, \frac{5\pi}{4}$

③ 算 $\cos(\theta) = 0$ 的 θ 值: $\cos(\theta) = 0$ $\frac{\pi}{2}$ $\frac{\pi}{c}$ Q1, Q4

$0 \leq \theta \leq 2\pi$, $\theta = \frac{\pi}{2}, \frac{3\pi}{2}$

④ Conclude: $\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{4}, \frac{5\pi}{4}$

○ ① 转换 $\sin^2(\alpha) = \cos^2(\alpha)$: $\tan^2(\alpha) = 1$

$\tan(\alpha) = \pm 1$

② 分别算 $\tan(\alpha) = \pm 1$: when $\tan(\alpha) = 1$, $\frac{\pi}{4}$, $\frac{\pi}{c}$

$-\pi \leq \alpha \leq \pi$, $\alpha = \frac{\pi}{4}, -\frac{3\pi}{4}$

when $\tan(\alpha) = -1$, $\frac{\pi}{4}$ $\frac{\pi}{c}$

$-\pi \leq \alpha \leq \pi$, $\alpha = \frac{3\pi}{4}, -\frac{\pi}{4}$

③ Thus, $\alpha = -\frac{3\pi}{4}, -\frac{\pi}{4}, \frac{\pi}{4}, \frac{3\pi}{4}$

○ Type: solve $(2\sin(2x + \frac{\pi}{4})) = \sqrt{2}$ $|\ -\pi \leq x \leq \pi, x)$
 math3

用 standard Real Rad

↑
因为 π 在算式里。

$\alpha = 0, -\pi, \pi, -\frac{3\pi}{4}, \frac{\pi}{4}$

数 General Solutions of Trigonometric Equations 2.4

↑↑↓↑↑↑↑↑↑↑↑↑↑↑↑↑↑

Q1 Find the General solution of the equation $\sqrt{3} \tan(2\theta) + 1 = 0$

and then find all the solutions for $\theta \in [-\pi, \pi]$

$$\theta = \frac{\pi}{2} + \tan^{-1}(\alpha) \quad n \in \mathbb{Z}$$

① isolate $\tan(2\theta)$: $\tan(2\theta) = \frac{1}{\sqrt{3}} = -\frac{\sqrt{3}}{3}$

② identify the Basis angle: $-\frac{\sqrt{3}}{3} = -\frac{\pi}{6}$

错误想法: 去找其他的 $\tan(\theta) = -\frac{\sqrt{3}}{3}$

如 $\frac{5\pi}{6}$, 其实并不用。因为 we always pick the smaller angle

③ Apply the general solution formula:

$\theta = n\pi + \tan^{-1}(a) \quad n \in \mathbb{Z}$

如果是 sin 或 cos, 则有两个 formula

易错:

题目是 2θ 哦

$2\theta = n\pi + (-\frac{\pi}{6}) \quad n \in \mathbb{Z}$ ← 这个不能忘!

$2\theta = \frac{6n\pi}{6} - \frac{\pi}{6}$

$2\theta = \frac{\pi(6n-1)}{6}$

$\theta = \frac{\pi(6n-1)}{12}, \quad n \in \mathbb{Z}$

state the solution with the given domain

④ For $\theta \in [-\pi, \pi]$, 整几个整数塞进公式里:

* when $n = -2$, $\theta = -\frac{13\pi}{12}$ ← rejected as $\theta \in [-\pi, \pi]$

$n = -1$, $\theta = -\frac{7\pi}{12}$

$n = 0$, $\theta = -\frac{\pi}{12}$

$n = 1$, $\theta = \frac{5\pi}{12}$

$n = 2$, $\theta = \frac{11\pi}{12}$

⑤ $\therefore \theta = -\frac{7\pi}{12}, -\frac{\pi}{12}, \frac{5\pi}{12}, \frac{11\pi}{12}$

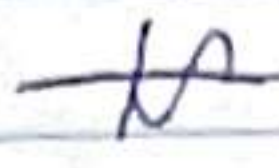
中数 Circular function 2.5

易错: 看 $\sin(x)$ 图猜 formula 时, 注意 Period, 别把 y -值看成 period.

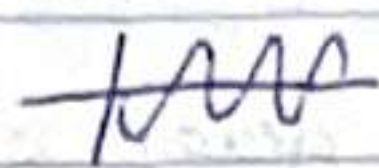
↑↑ | ↑↑ | ↑↑ | ↑↑ (↑↑) 24/10/2021 T ↑↑ | ↑↑ | ↑↑ | ↑↑

Q1 Sketch $y = 2 - 4\sin(3x)$ $0 \leq x \leq 2\pi$
不用改成 $0 \leq 3x \leq 2\pi$
因为给的是 π -axis 上的限制

① Mean position $\div y=2$ Range $[-2, 6]$

Amplitude $=4$ shape of sine graph 

Period $\frac{2\pi}{3}$ 看有几个 period: $2\pi \div \frac{2\pi}{3} = 3$

 对画图的把控很重要!

End point $y = 2 - 4\sin(6\pi)$

$=2$ $(2\pi, 2)$

x-intercepts: $y=0$

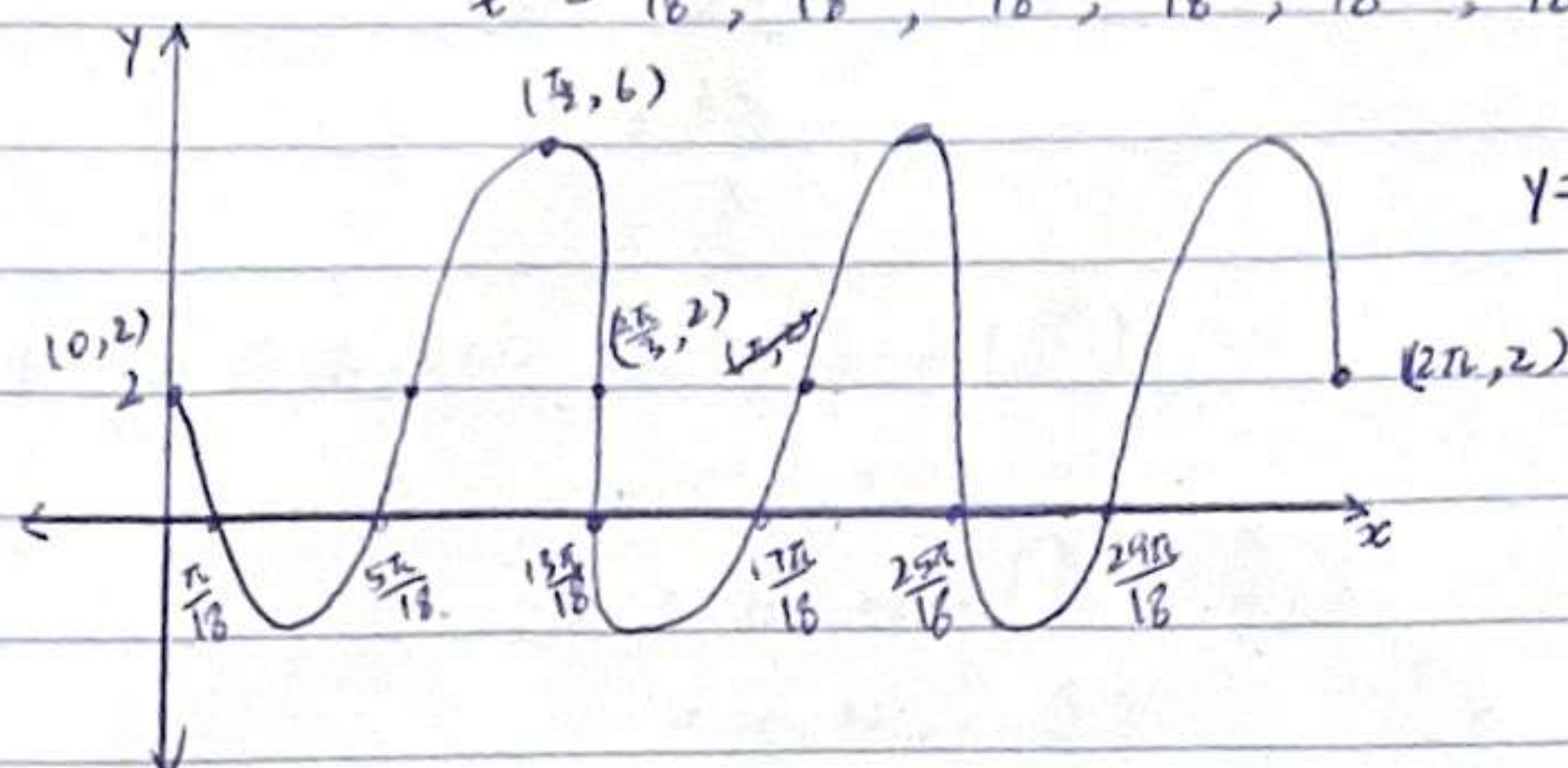
$$2 - 4\sin(3x) = 0$$

$$\sin(3x) = \frac{1}{2}, \quad \frac{\pi}{6} \quad 0 \leq 3x \leq 6\pi \quad \begin{array}{c} S \\ A \end{array}$$

$$3x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}, \frac{25\pi}{6}, \frac{29\pi}{6}$$

$$x = \frac{\pi}{18}, \frac{5\pi}{18}, \frac{13\pi}{18}, \frac{17\pi}{18}, \frac{25\pi}{18}, \frac{29\pi}{18}$$

②



$$y = 2 - 4\sin(3x), \quad 0 \leq x \leq 2\pi$$

2.6 中数

The tangent function

↑↑ | ↑↑ | ↑↑ | ↑↑ | (↑↑) | ↑↑ | ↑↑ | ↑↑ | ↓

Q1 The graph of $y = a \tan(\pi x)$ has domain $(-\frac{\pi}{3}, \frac{\pi}{3})$ with asymptote at $x = -\frac{\pi}{3}$, $x = \frac{\pi}{3}$ only, it also passes through $(0, 0)$ $(-\frac{\pi}{6}, -\frac{1}{2})$ Find the equation

○ 错误点: 混淆了求 period 和求 asymptote 的方法, ~~实际上两种方法都能求出 n~~

$$\text{Period} = \frac{\pi}{n} \leftarrow \text{原 period} \quad \text{Asymptote } x = \frac{\pi}{n} \leftarrow \text{原 asymptote} \\ \text{for this graph}$$

正确: ~~将 π 代入 $x = \frac{\pi}{n}$~~

① Period: $\frac{\pi}{n} = \text{period}$

此 graph period 等于两个 asymptote 的间隔

$$\frac{\pi}{3} + \frac{\pi}{3} = \frac{2\pi}{3}$$

Thus, 求 n : $\frac{\pi}{n} = \frac{2\pi}{3}$

$$3\pi = 2\pi n$$

$$n = \frac{3\pi}{2\pi}$$

$$= \frac{3}{2}$$

② 求出 n 后找 a : $y = a \tan\left(\frac{3x}{2}\right)$

use $\left(-\frac{\pi}{6}, -\frac{1}{2}\right)$

$$-\frac{1}{2} = a \tan\left(\frac{3}{2}x - \frac{\pi}{6}\right)$$

$$= a \tan\left(-\frac{\pi}{4}\right)$$

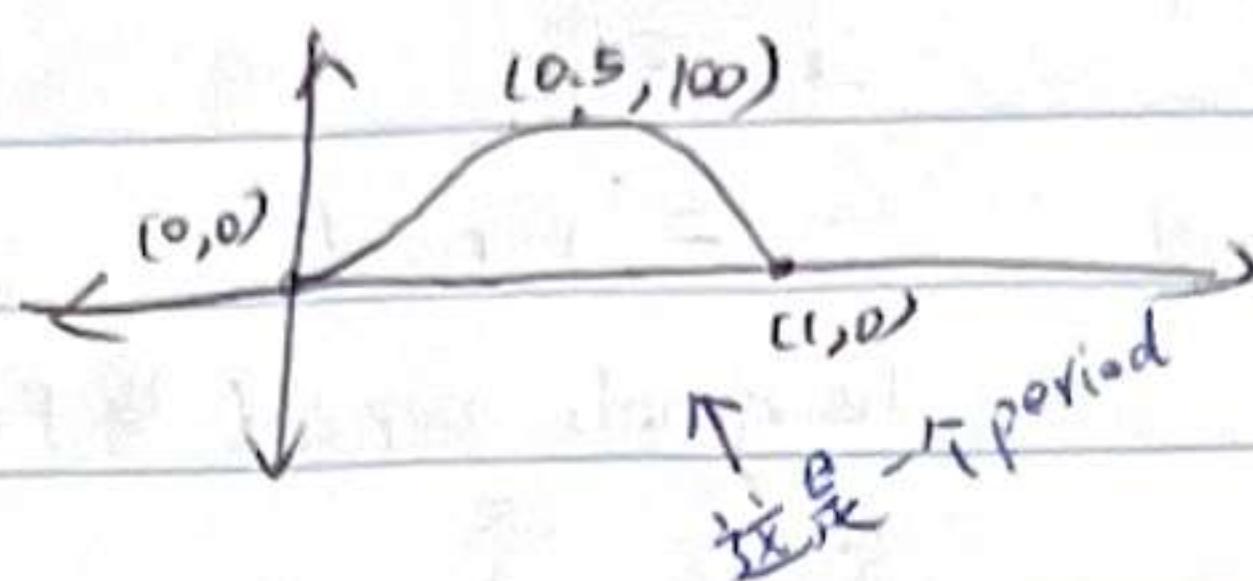
$$-\frac{1}{2} = a \times -1$$

$$a = \frac{1}{2}$$

③ Conclude: $y = \frac{1}{2} \tan\left(\frac{3x}{2}\right)$

中数 2.1 Modelling and Applications.

Q1 Determine the graph formula: $y = a \cos(\pi t) + c$
find a, n, c



中数 3.2 Composite functions 复合函数

Q1 $f(x) = x^2$, $g(x) = \sqrt{x}$, find rule of $g(f(x))$

Q2. $g(x) = \frac{1}{(x-3)^2} - 2$, $f(x) = \sqrt{x}$
(a) Prove $f(g(x))$ is not defined

(b) Restrict domain of $g(x)$ to obtain $g_1(x)$ such that $f(g_1(x))$ exists.

○ 易错: 从 shape 看正反

$$\text{Mean} : y = 50$$

$$\text{Amplitude } (100 - 0) \div 2 = 50$$

$$\text{Period} = 1$$

$$a = 50 \quad c = 50$$

$$\frac{2\pi}{n} = 1 \quad n = 2\pi$$

$$h = 50 \cos(2\pi t) + 50$$

是负 cos 图, thus, $h = -50 \cos(2\pi t) + 50$

○ $g(f(x)) = g(x^2)$

$$= \sqrt{x^2}$$

$= |x|$ or $\pm x$ 因为 x 可能是负数也可能是正数

○ (a) Test if range of $g(x) \subseteq \text{domain of } f(x)$

① $g(x)$: Domain $\mathbb{R} \setminus \{3\}$ $f(x)$: Domain $[0, \infty)$

Range ~~$(-2, \infty)$~~ $(-2, \infty)$ Range $[0, \infty)$

② range of $g(x) \not\subseteq \text{domain of } f(x)$

$(-2, \infty) \not\subseteq [0, \infty)$ Thus, $f(g(x))$ not exist.

(b) range of $g(x) \subseteq \text{domain of } f(x)$

range of $g_1(x) = [0, \infty)$ domain of $f(x) = [0, \infty)$

当 $g(x)$ range $= [0, \infty)$, 求 $g(x)$ 的 domain by set $g(x) \leq 0$

$$\frac{1}{(x-3)^2} - 2 \leq 0$$

CAS: $x \in (-\infty, -\frac{1}{\sqrt{2}} + 3] \cup [\frac{1}{\sqrt{2}} + 3, \infty)$

$$\frac{1}{2} \leq (x-3)^2$$

$$\frac{1}{2} \leq [+(x-3)]^2 \cup \frac{1}{2} \leq [-(x-3)]^2$$

$$\frac{1}{\sqrt{2}} \leq x-3 \quad -\frac{1}{\sqrt{2}} \geq x-3$$

$$x \geq 3 + \frac{1}{\sqrt{2}} \cup x \leq 3 - \frac{1}{\sqrt{2}}$$

中数 3.3 Functional Equations

↑↑↑↑↑↑↑↑↑↑ (15/05/2022)

Q1. if $f(x) = e^{2x}$, state functional equation for $f(x+y)$ and $f(x-y)$

$$0 \quad f(x+y) : e^{2(x+y)} = e^{2x+2y} = e^{2x} \times e^{2y} = f(x)f(y)$$

$$f(x-y) : e^{2(x-y)} = e^{2x-2y} = \frac{e^{2x}}{e^{2y}} = \frac{f(x)}{f(y)}$$

中数 3.4 Transformations

Q1 Describe the transformation

$$y = x^3 \rightarrow y = -2(1-x)^3 + 1$$

从左往右读

Q2. The graph $y = x^2$ undergoes

- reflection in y -axis
- translation 4 units right
- dilated of factor 2 from y -axis
- translation 3 units down
- dilation of factor $\frac{1}{3}$ from x -axis

Q3. State the transformation

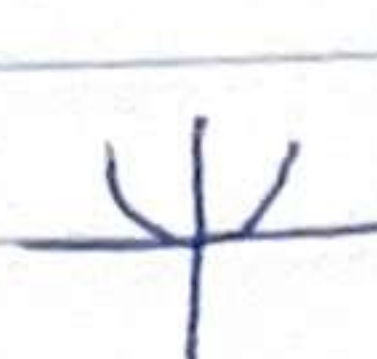
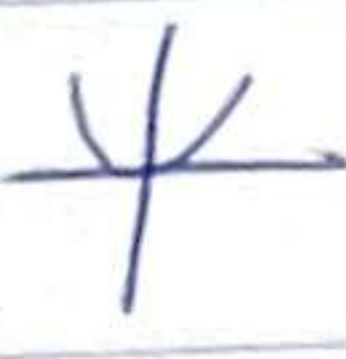
$$y = -2(3x-1)^2 + 5 \rightarrow y = (x+2)^2 - 1$$

从左到右

○ ★ 凡是 x 前是负号, 加多少往右移多少

- reflected in the x -axis
- dilated by factor of 2 from x -axis
- reflected in the y -axis
- translated 1 unit to the right
- translated 1 unit upward.

○ • ① $y = (-x)^2$
 $= x^2$

 reflect in y -axis 还是 

② $(x-4)^2$

③ $(\frac{x}{2}-4)^2$

④ $(\frac{x}{2}-4)^2 - 3$

⑤ $\frac{1}{3} [(\frac{x}{2}-4)^2 - 3]$
 $= \frac{1}{3} (\frac{x}{2}-4)^2 - 1$

○ • reflected in x -axis

- related.
- dilated by factor of $\frac{1}{2}$ from x -axis
 - dilated by factor of 3 from y -axis
 - translated 3 units to the left
 - ★ • translated ~~to~~ $\frac{3}{2}$ units up.
易错点

$y = 2(3x-1)^2 - 5$

$y = \frac{1}{2} [2(3x-1)^2 - 5] = (3x-1)^2 - \frac{5}{2}$

$y = (\frac{3x}{3}-1)^2 - \frac{5}{2}$

$y = (x+2)^2 - \frac{5}{2}$

$y = (x+2)^2 - 1$

整个 graph
因为负号 applied to

3.6 中数 Inverse graph
Q1 $y = 2x^2 - 12x + 13$, domain $(-\infty, a]$ find value of a so
that inverse of y is a function

3.7 中数 Inverse function.

Q1 Find point of intersection between $f(x) = 2\sqrt{x+2}$ and its $f^{-1}(x)$

$$0 \quad 2(x^2 - 6x + \frac{13}{2})$$

$$y = 2(x^2 - 6x + (3)^2 - (3)^2 + \frac{13}{2})$$

$$y = 2(x-3)^2 - 5 \quad \leftarrow \text{TP form} \quad \text{TP } (3, 5)$$

$$\cdot \text{ so } a = 3$$

TP coordinates $(-\frac{b}{2a}, f(-\frac{b}{2a}))$

$\downarrow$

$\frac{12}{4} = 3$

$a = 3$

Q.

$$\text{Solve } 2\sqrt{x+2} = x$$

$$4(x+2) = x^2$$

$$x^2 - 4x - 8 = 0$$

$$x^2 - 4x + 4 - 4 - 8 = 0$$

$$(x-2)^2 - (\sqrt{12})^2 = 0$$

$$(x-2+2\sqrt{3})(x-2-2\sqrt{3}) = 0$$

$$x = 2+2\sqrt{3}, 2-2\sqrt{3}$$

$$\text{Intersect Point } (2+2\sqrt{3}, 2+2\sqrt{3}) \quad (2-2\sqrt{3}, 2-2\sqrt{3})$$

rejected since
domain of $f(x)$
is $[2, \infty)$
 $[0, \infty)$

中数 (class practice)

↑↑ | ↑↑ | ↑↑ | ↑↑ | ↑↑ | ↑↑ | ↑↑ | ↑↑

Q1 $g(x) = x + 2$ $f(x) = x^2 - 4$

if h is the composite function given by: $h: [-5, -1) \rightarrow \mathbb{R}$

$h(x) = f(g(x))$, find

the range of h is ?

↑↑ | ↓↑ | ↑↑ | ↑↑ | ↑↑ | ↑↑ | ↑↑ | ↑↑

Related to inverse function

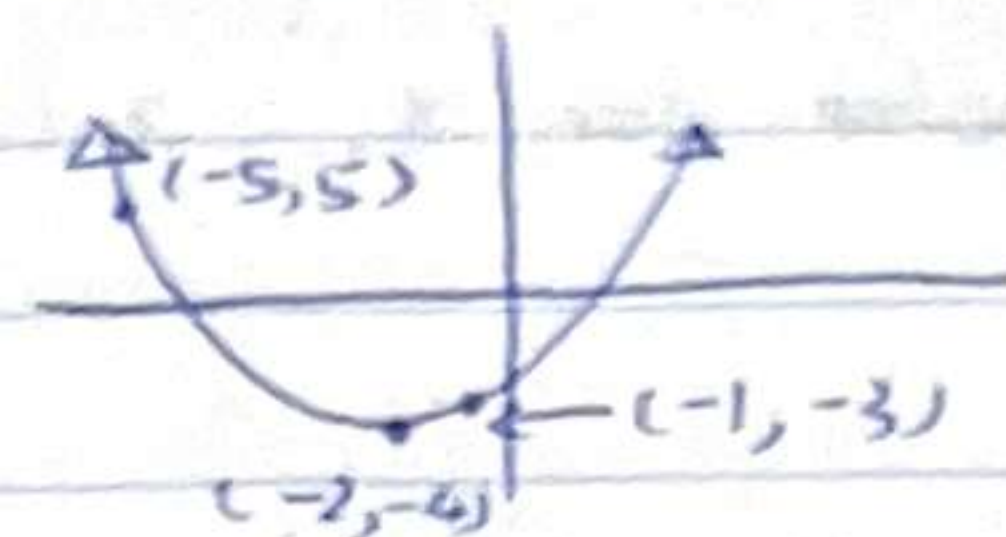
Q2 Make y the subject: ~~$y = x^2 - 2y$~~ $x = y^2 - 2y$

最后记得根据 inverse 的 domain 挑选 sol.

○ ① 用 sketch 确保不出错: $f(x+2) = (x+2)^2 - 4$

TP $(-2, -4)$

Range



~~$h(-5) = 5$~~ ~~$h(-1) = -3$~~

Thus, range = $[-4, 5]$

○ $x = y^2 - 2y + 1 - 1$

$x = (y-1)^2 - 1$

$(y-1)^2 = x+1$

$y-1 = \pm\sqrt{x+1}$

$y = 1 \pm \sqrt{x+1}$

4.4 中数 Indical equations

Q1 Solve for x , $2(5^x) - 12 = -\frac{10}{5^x}$

Q2 Solve for x , $4^{x+1} = 3^{1-x}$

$$0 \quad [2(5^x) - 12 = -\frac{10}{5^x}] \times 5^x, \quad 2(5^x)^2 - 12 \cdot 5^x = -10$$

$$\text{set } a = 5^x$$

$$2a^2 - 12a + 10 = 0$$

$$a^2 - 6a + 5 = 0$$

$$5 < -1$$

$$(a-1)(a-5) = 0$$

$$5^x = 5$$

$$x = 1$$

$$5^x = 1 \quad || \quad 5^0$$

$$x = 0$$

$$x = 0, 1$$

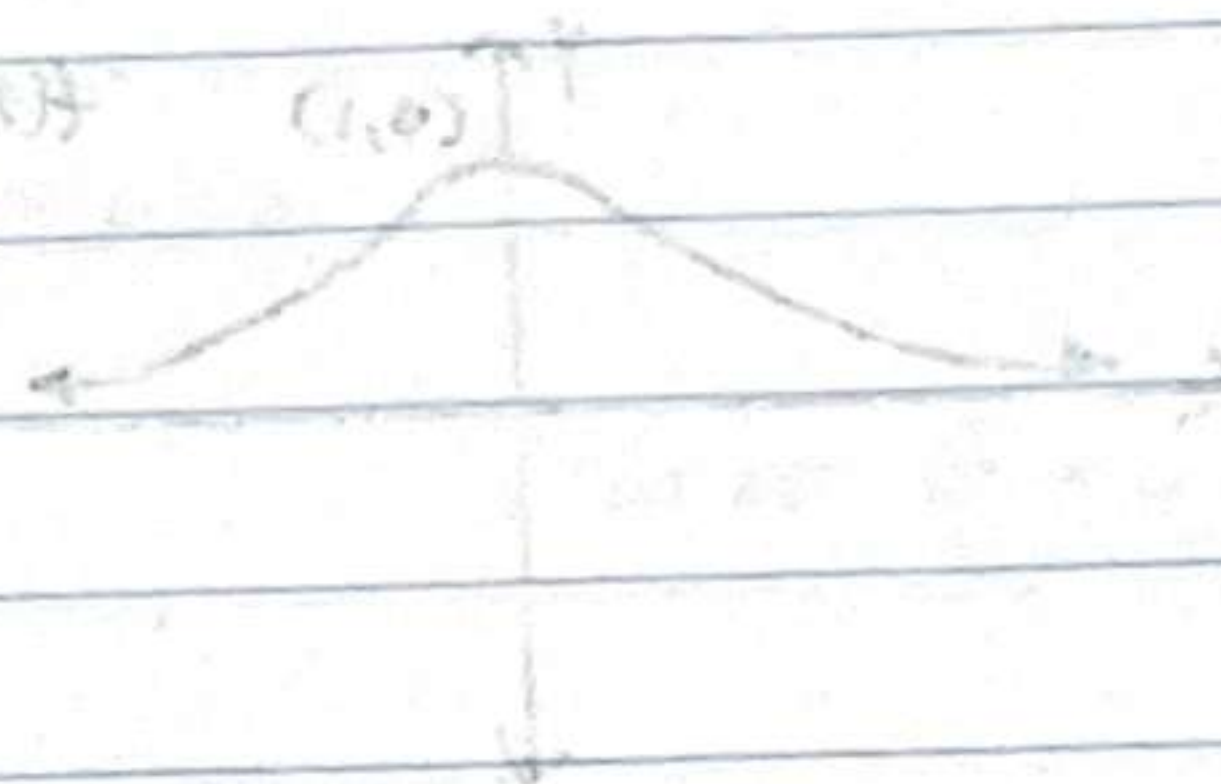
$$0 \quad \log_e(4^{x+1}) = \log_e(3^{1-x})$$

$$(x+1) \log_e(4) = (1-x) \log_e(3)$$

$$x \log_e(4) + \log_e(4) = \log_e(3) - x \log_e(3)$$

$$x \log_e(12) = \log_e\left(\frac{3}{4}\right)$$

$$\text{Ans. } x = \frac{\log_e\left(\frac{3}{4}\right)}{\log_e(12)}$$



数 14.4 Confidence intervals

↑↑↑↑↑

Q1 30 people surveyed, 78% like breakfast.

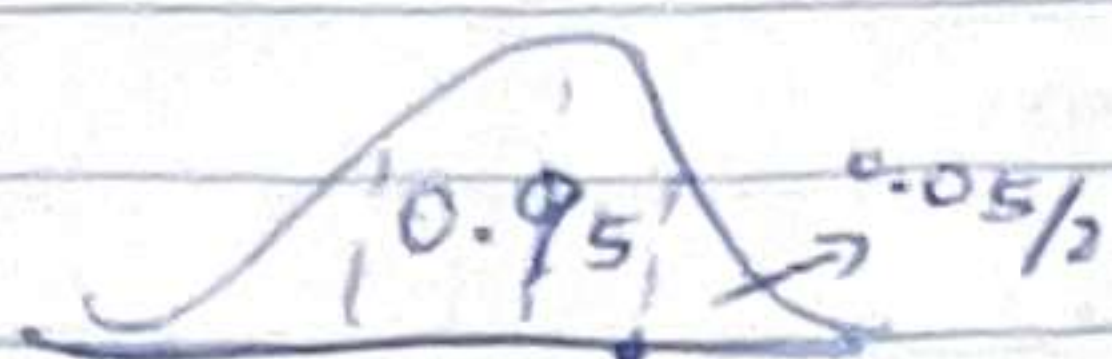
Estimate the $\hat{p}$ that like breakfast

Find a 95% confidence interval for the estimate, to 2 decimal

↑↑↑↑↑

Q2 With belief that 65% of people have brown hair. 250 people are surveyed. Applying the normal approximation. Find the probability that less than 60% of people in the sample have brown hair. 四位小数

Find z :



at $0.95 + 0.025 = 0.975$

Inv Norm CDF(z , 0.975, 1, 0)

$z \approx 1.96$

Find interval:

$$\hat{p} = 0.78, n = 30$$

$$\hat{p} - z \sigma_{\hat{p}} = 0.78 - 1.96 \cdot \sqrt{\frac{0.78 \times 0.22}{30}} \approx 0.6318$$

$$\hat{p} + z \sigma_{\hat{p}} \approx 0.9282$$

Confidence interval = $(0.6318, 0.9282)$

0

$$n = 250, \hat{p} = 0.65$$

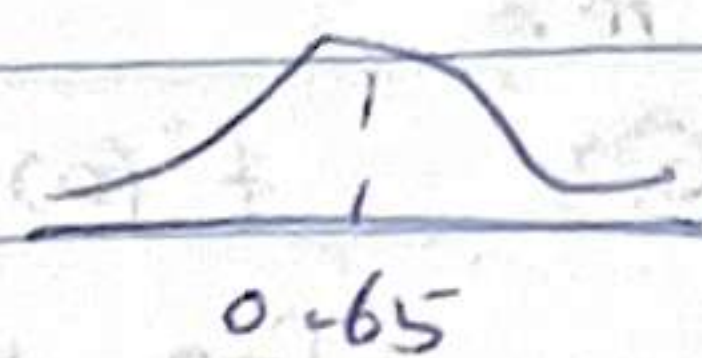
Approximate the distribution of $\hat{p}$ as a normal curve

$$\mu = p = 0.65$$

$$\sigma = \sqrt{\frac{0.65 \times 0.35}{250}} = 0.03$$

$$\hat{p} \sim \text{Normal}(0.65, 0.03^2)$$

$$Pr(\hat{p} < 0.6) \approx 0.0487$$



VCAA 2018 exam 2

Multiple choice $\uparrow\uparrow$ Q1 $y = \tan(ax)$ has a vertical asymptote $x = 3\pi$ and has exactly one x -intercept in the region $(0, 3\pi)$. Find the value of a .

$\uparrow\uparrow$ Q2 The turning point of the parabola $y = x^2 - 2bx + 1$ is closest to the origin when value(s) of $b = ?$ ~~$b \neq 0$~~

$\uparrow\uparrow$ Q3 $f'(0) = 4$

The transformation maps $f(x)$ to $g(x)$
now, $g'(0) = -1$

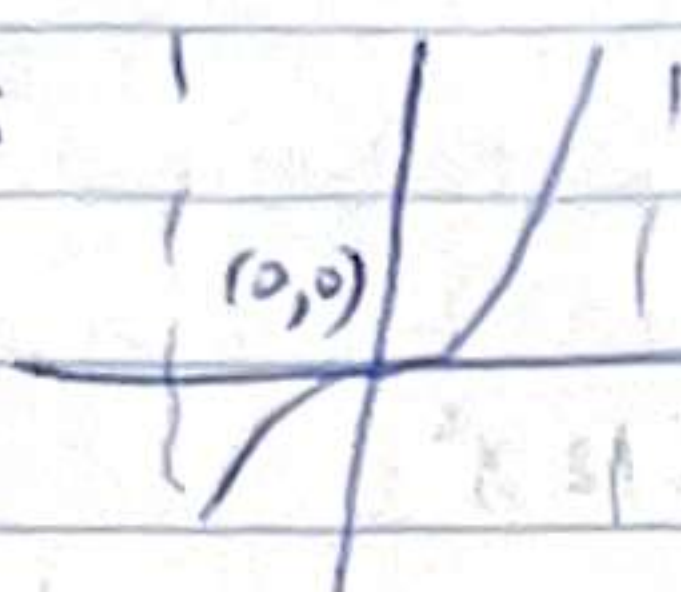
The transformation T could be given by:

A. $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} -2 & 0 \\ 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

B. $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

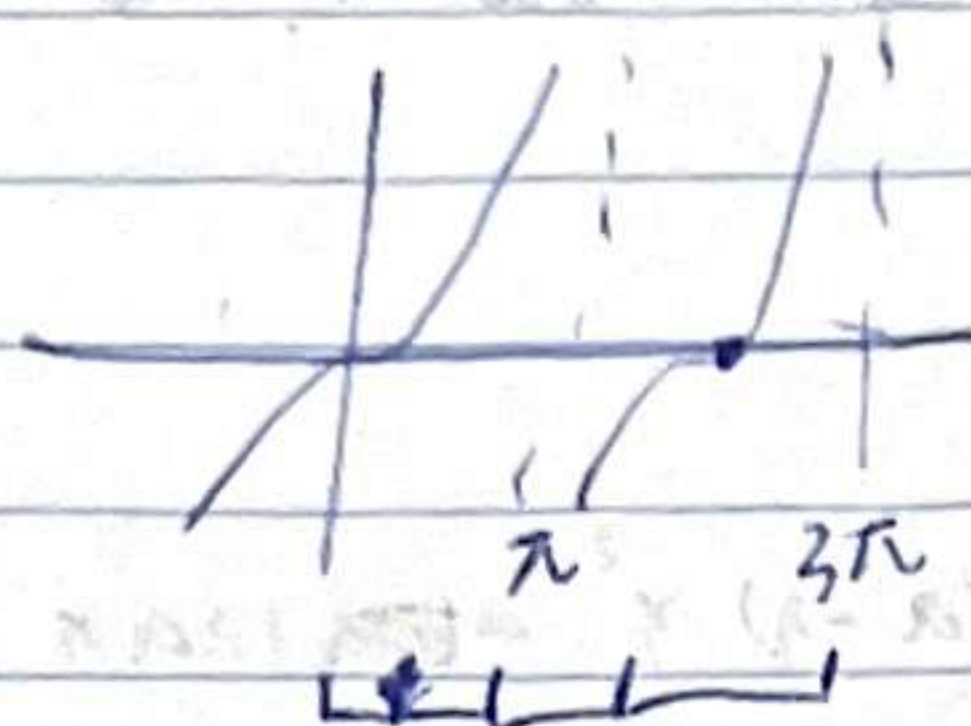
C. $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} -\frac{1}{2} & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

scenario 1:



但 $(0, 3\pi)$ 有一个 asymptote
而这个没有

scenario 2:



这下有了

$3\pi \div 3 = \pi$ 半 $\tan \sqrt{\quad}$ 是 π

$$\text{然后 } \frac{\pi}{2} = ax$$

$$x = \frac{\pi}{2a} = \pi$$

$$a = \frac{1}{2}$$

TP x-coordinate $x = \frac{-b}{2a} = \frac{-(-2b)}{2} = b$

when $x = b$, $y = 1 - b^2$

Distance = $\sqrt{b^2 + (1 - b^2)^2}$

$\frac{db}{db} = 0$, $b = \pm \frac{1}{\sqrt{2}}$

① dilated by factor -2 from y-axis
 $\frac{4}{-2} = -2$

② dilated by factor $\frac{1}{2}$ from x-axis
 $-2 \times \frac{1}{2} = -1$

ans = A

VCAA 2018 exam 2 应用题

Q1 ↑↑

用CAS:

$$f(x) = 3x^4 + 4x^3 - 12x^2$$

Find ~~the~~ tangent equa at $x = -\frac{1}{3}$

Q2

$$p(x) = 3x^4 + 4x^3 + 6(a-2)x^2 - 12ax + a^2, a \in \mathbb{R}$$

↑↑

(a) Find all solutions to $p'(x) = 0$ in terms of a .

↑↑

(b) Find values of a for $p(x)$ has only one stationary point

① Interactive / calculation / line / tanLine /

Expression : $f(x)$

Point : $-\frac{1}{3}$

$$\text{Ans : } y = \frac{80}{9}x + \frac{24}{27}$$

① (a) $p'(x) = 0$, $x = 1$, $\pm \sqrt{1-a}$ -1

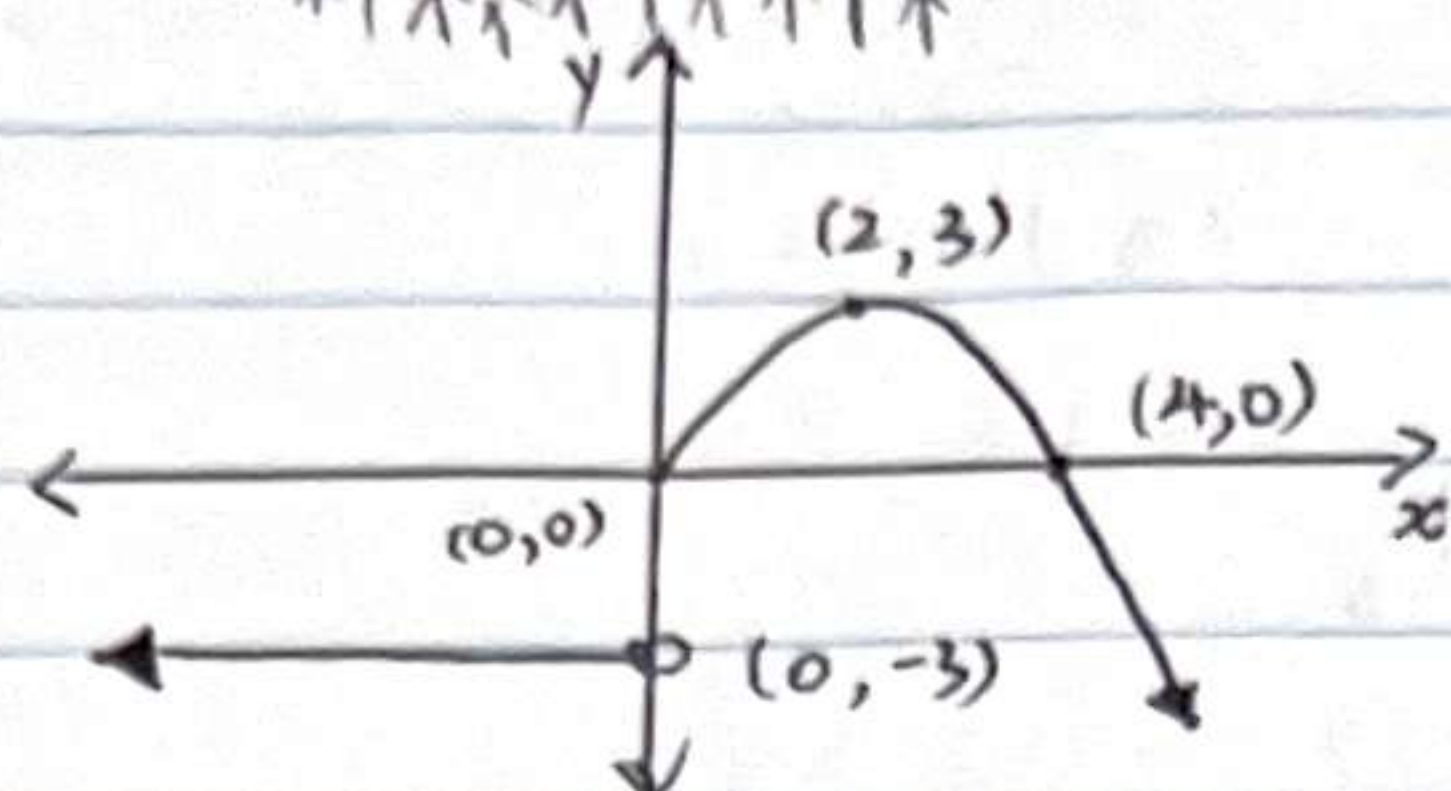
(b) $1-a < 0$

~~$a < 1$~~

$a > 1$

5.2 Review of differentiation

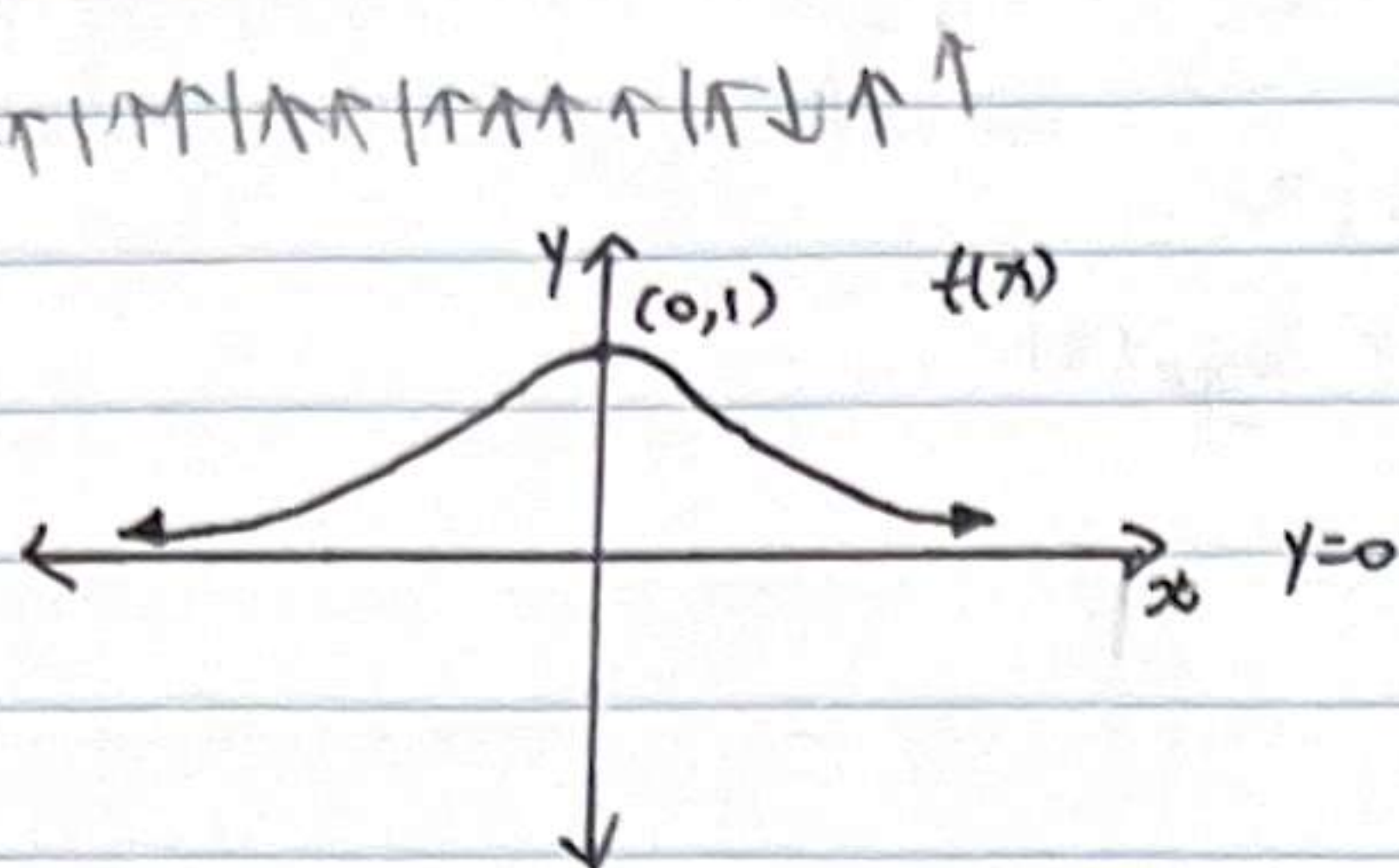
Q1



(a) State domain of $f'(x)$

(b) Sketch $f'(x)$ and intercepts

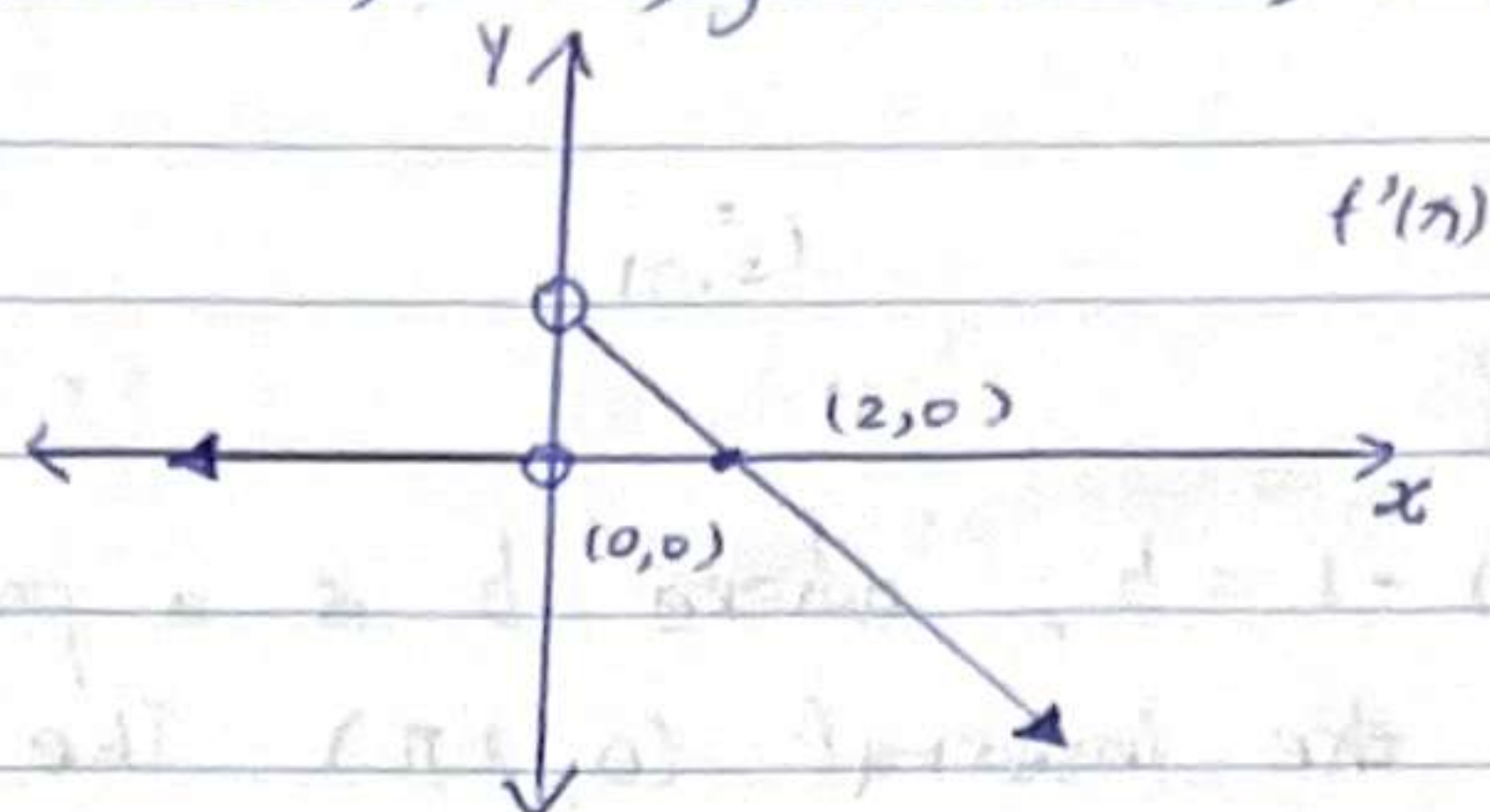
Q2.



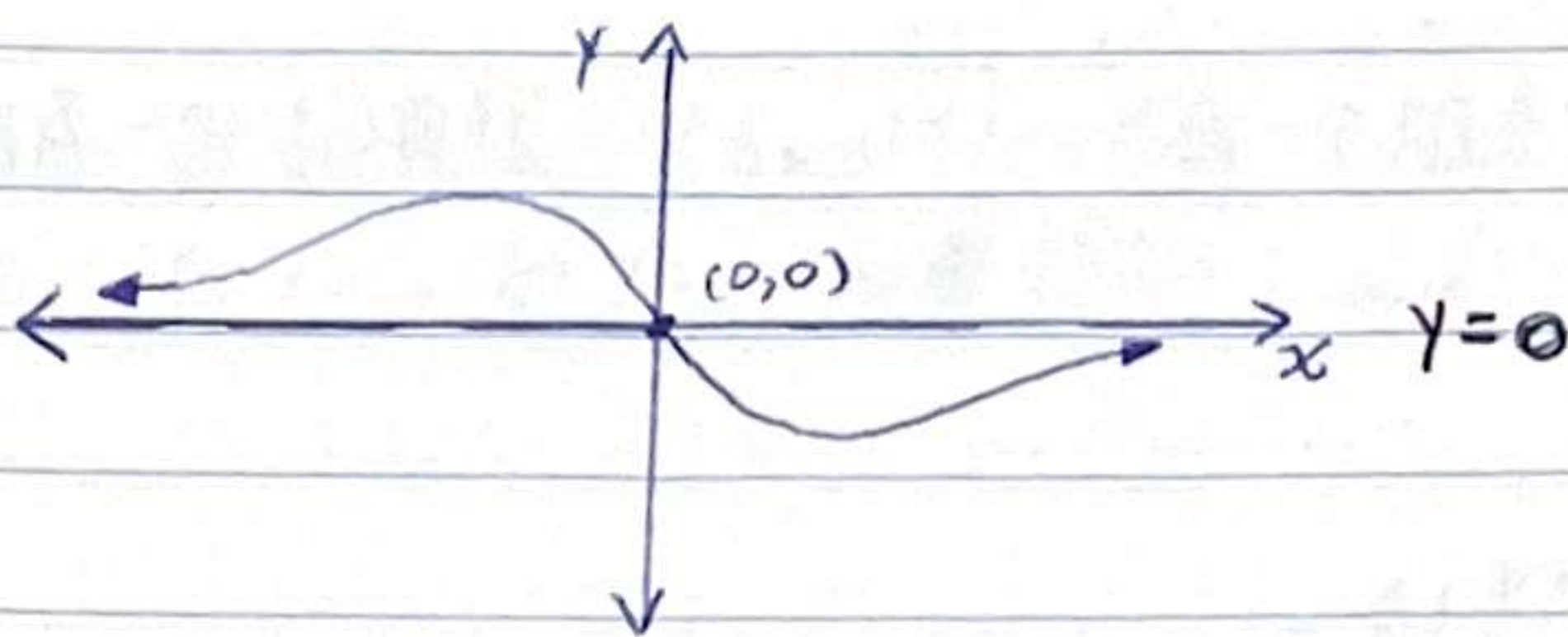
(a) State domain of $f'(x)$

(b) Sketch $f'(x)$

- Domain
- (a) $\mathbb{R} \setminus \{0\} \therefore$ discontinuity at $x=0$
- (b) IP at $x=2$ for $f(x)$, $f'(x)$ x -ins at $x=2$
- $f(x)$ x 值 $\in (-\infty, 0)$, gradient $= 0$, 因为是平线
- $f(x)$ x 值 $\in (0, 2)$, gradient > 0 , 因为在上升
- $f(x)$ x 值 $\in (2, \infty)$, gradient < 0 , 因为在下降



- Domain
- (a) $\mathbb{R}$ 因为 Graph smooth and continue
- (b)



x -ins at $x=0$
因为 IP 在 $x=0$
f'(x) 的

* 知识点: 凡是 $f(x)$ 有 Horizontal asymptote 的, 其 $f'(x)$ 都有 $y=0$ 的 Asymptote 因为 graph of $f(x)$ 接近平缓。

5.3 中数 Differentiation of exponential functions

Q1. Differentiate $5e^{-4x} + 2e$

中数 Quiz 1

Q1. The question $3\sin(x) - 1 = b$, where b is a positive real number, has one solution in the interval $(0, 2\pi)$. The value of b is:

A. 2 两者相交点

B. 0.2

C. 3

D. 5

E. 6

Q2. The curve with equation $y = a\sin(bx) + c$ is mapped onto the curve with equation $y = 2\sin(x + \frac{\pi}{2}) - 3$ by the Transformation:

$$T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} \frac{\pi}{2} \\ 3 \end{bmatrix}$$

Find value of a, b, c

IAS Active

0 $-20e^{-4x}$

$e^x < 1$ 的 $f'(x)$ 是 0

① ① Sketch $3 \sin(x) - 1$:

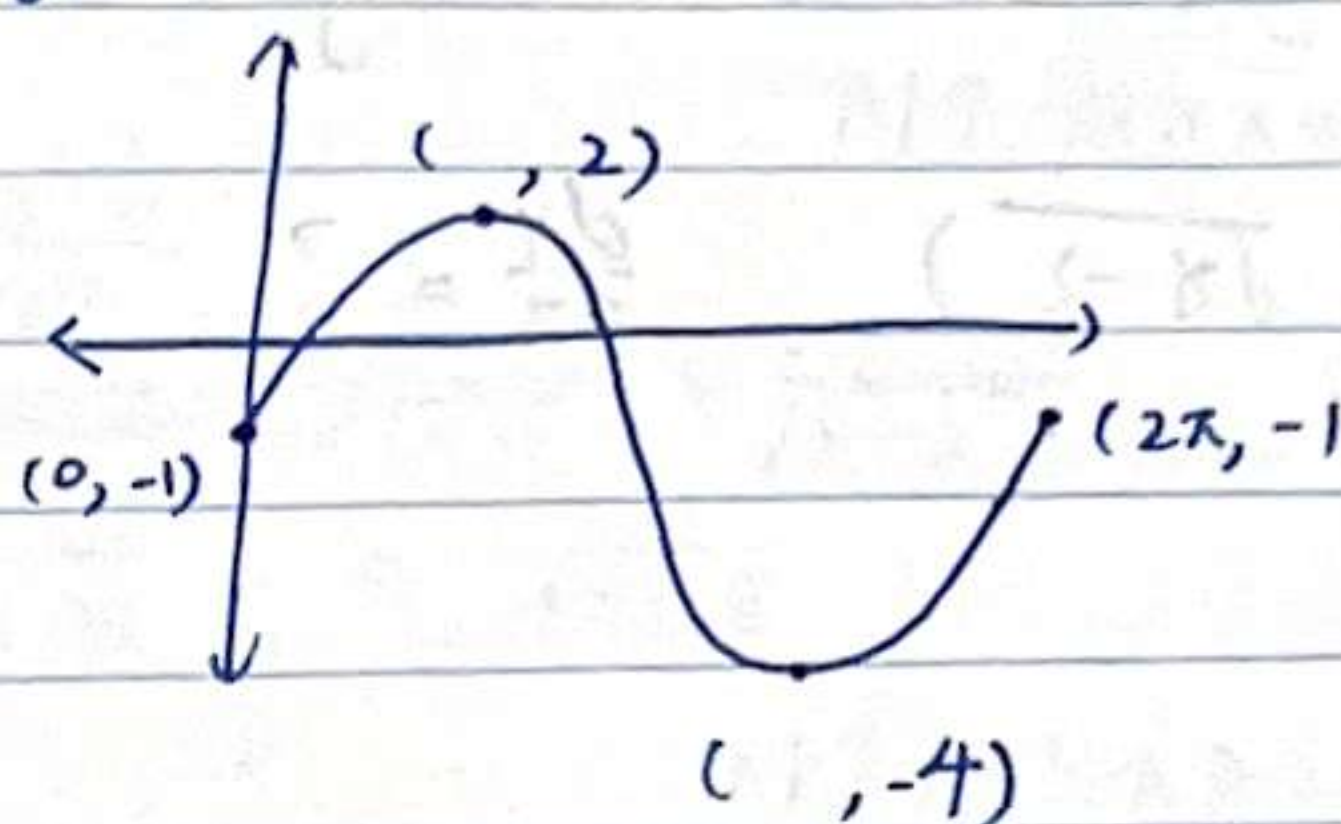
Amplitude: 3

Mean: -1

Range: $[-3, 3] - 1$

Period: 2π

$[-4, 2]$



② 将 b 看成一条横直线, 与 $3 \sin(x) - 1$ 只有一个相交点.

因此只有 $y = 2$ 和 $y = -4$ 符合, and b 是正数所以只能是 2.

A.

① ① $x' = \frac{1}{2}x + \frac{\pi}{2}$
 $y' = -y + 3$

② 理解 $y = a \sin(bx) + c \Rightarrow 2 \sin(x + \frac{\pi}{2}) - 3$
 $\therefore$ 此 equa 里的 y 和 x 是 y' 和 x' 后妈后爸

原配驾到:

③ $-y + 3 = 2 \sin(\frac{1}{2}x + \frac{\pi}{2} + \frac{\pi}{2}) - 3$

$-y = 2 \sin(\frac{1}{2}x + \pi) - 6$

$y = -2 \sin(\frac{1}{2}x + \pi) + 6$

(As: $\sin(\frac{1}{2}x + \pi)$ can be simplify to $-\sin(\frac{x}{2})$)

$\therefore y = 2 \sin(\frac{x}{2}) + 6$

$a = 2, b = \frac{1}{2}, c = 6$

5.5

Differentiation of trigonometric function.

Q1. Find $\frac{dy}{dx}$ for $y = \sin(\frac{2}{3}x^\circ)$

↳ to rad: $\times \frac{\pi}{180^\circ}$

9.2 Derivative of $f(x) = \log_e(x)$

Q1 $\frac{1}{2} \log_e (\sqrt{x-2})$, $\frac{dy}{dx} = ?$

Q2. $y = \log_e \left(\frac{x+3}{x+1} \right)$, Find $\frac{dy}{dx}$ (不用 expand 分母)

别干傻事,让事情复杂谓一

~~ITAN~~ Transformation

$y = (\frac{x}{2} - 3)^4$ move 2 units to right, state transformed

$$\textcircled{1} \quad y = \sin(2 \cdot x \cdot \frac{\pi}{180})$$

$$y = \sin\left(\frac{\pi x}{90}\right), \quad \frac{dy}{dx} = \frac{\pi}{90} \cos\left(\frac{\pi x}{90}\right)$$

$$\textcircled{2} \quad y = \frac{1}{2} \log_e(\sqrt{x-2}), \quad g(x) = \sqrt{x-2} \quad g'(x) = \frac{1}{2} (x-2)^{-\frac{1}{2}} \\ = \frac{1}{2\sqrt{x-2}}$$

$$\frac{dy}{dx} = \frac{1}{2} \cdot \frac{g'(x)}{g(x)}$$

$$= \frac{1}{2} \cdot \frac{1}{\sqrt{x-2}} \times \frac{1}{\sqrt{x-2}}$$

$$= \frac{1}{4(x-2)} = \frac{1}{4x-8}$$

$$\textcircled{3} \quad y = \log_e(x+3) - \log_e(x+1)$$

$$\frac{dy}{dx} = \frac{1}{x+3} - \frac{1}{x+1}$$

$$= \frac{(x+1) - (x+3)}{(x+3)(x+1)}$$

$$= \frac{-2}{(x+3)(x+1)}$$

$$\textcircled{4} \quad \text{要紧贴 } x$$

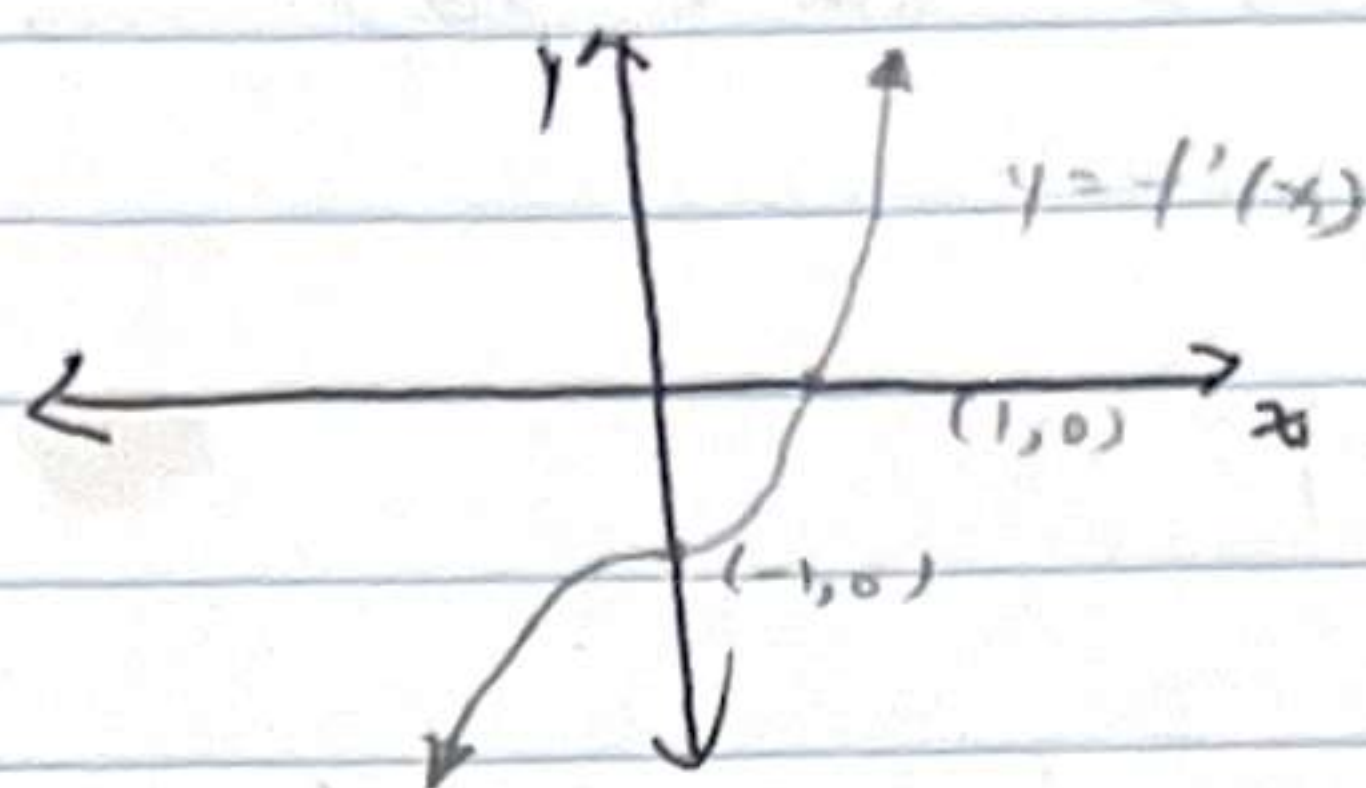
$$y = \left(\frac{(x-2)}{2} - 3\right)^4$$

$$= \left(\frac{x}{2} - 1 - 3\right)^4$$

$$= \left(\frac{x}{2} - 4\right)^4$$

7.4 Anti-diff curves

Q1 Sketch $f(x)$ if $f'(x)$ graph:



7.5

Application

The depth of water is given by:

$$h(t) = 2\sin\left(\frac{\pi t}{4}\right) + 3$$

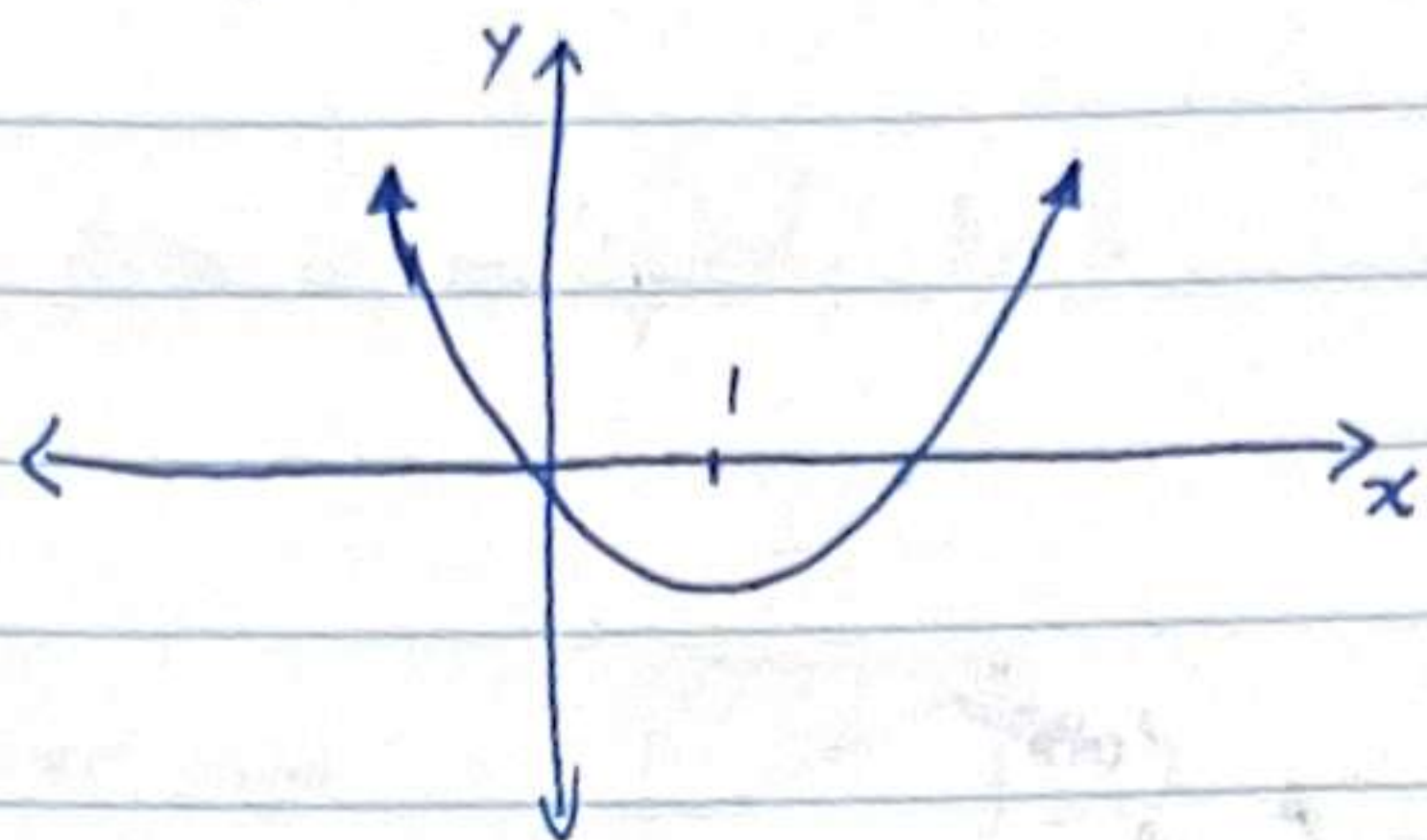
(a) What are the maximum and minimum depths of the water? in metres

(b) For how many hours a day is the water level 4 metres or more?

0 At $x=1$, SP

Before $x=1$, $f'(x) < 0$ ↘

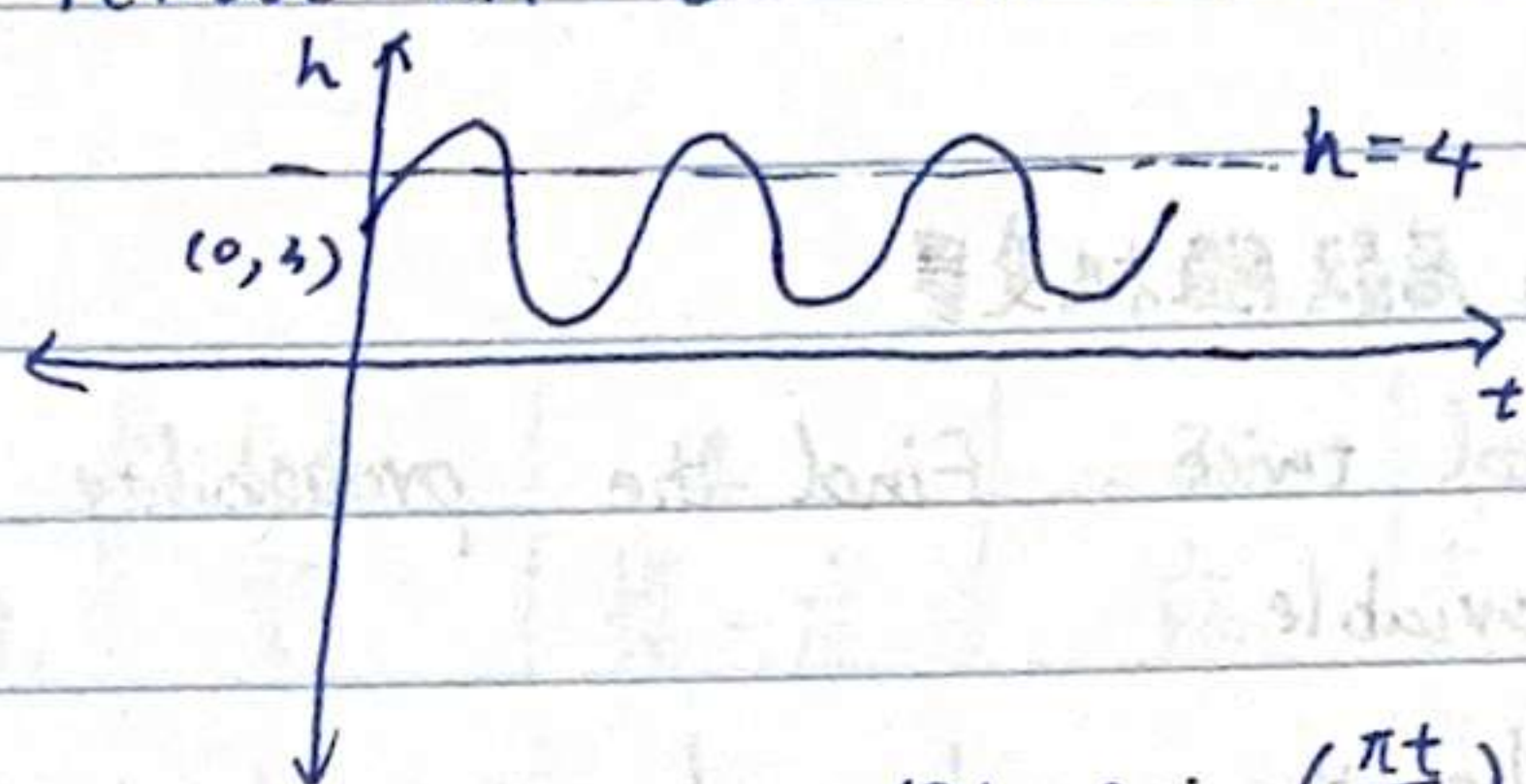
After $x=1$, $f'(x) > 0$ ↗



0 (a) Max : $2(1) + 3 = 5$ m

Min : $2(-1) + 3 = 1$ m

(b) Sketch : Amplitude : 2 Range $[1, 5]$ shape $\sin$
 Period : $\frac{2\pi}{\pi/4} = 8$ in 24 hours, 3 periods.



$0 \leq t \leq 24$

$0 < \pi t \leq 24\pi$

$0 \leq \frac{\pi t}{4} \leq 6\pi$

$\frac{36\pi}{6}$

set $2\sin(\frac{\pi t}{4}) + 3 = 4$

$\sin(\frac{\pi t}{4}) = \frac{1}{2} < \frac{\pi}{6}$ $\frac{5/4}{\pi/6}$

$\frac{\pi t}{4} = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}, \frac{25\pi}{6}, \frac{29\pi}{6}$

$t = \frac{2}{3}, \frac{10}{3}, \frac{26}{3}, \frac{34}{3}, \frac{50}{3}, \frac{58}{3}$

$h \geq 4$ when $t \in [\frac{2}{3}, \frac{10}{3}] \cup [\frac{26}{3}, \frac{34}{3}] \cup [\frac{50}{3}, \frac{58}{3}]$

hours = $(\frac{10}{3} - \frac{2}{3}) + (\frac{34}{3} - \frac{26}{3}) + (\frac{58}{3} - \frac{50}{3})$

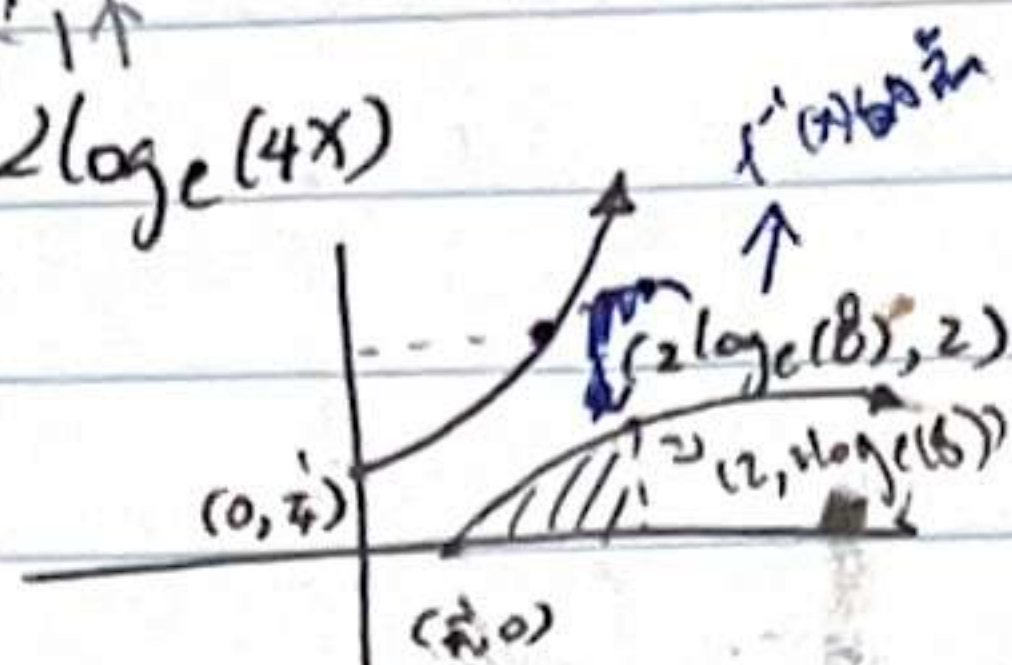
= $\frac{8}{3} + \frac{8}{3} + \frac{8}{3} = \frac{24}{3} = 8$ hours / day.

Class practice

Q1 Find $\int_0^3 f(3x) dx$, if $\int_0^9 f(x) dx = 5$

9.4 Application - integration

Q1 $f(x) = \frac{1}{4}e^{2 \log_e(4x)}$



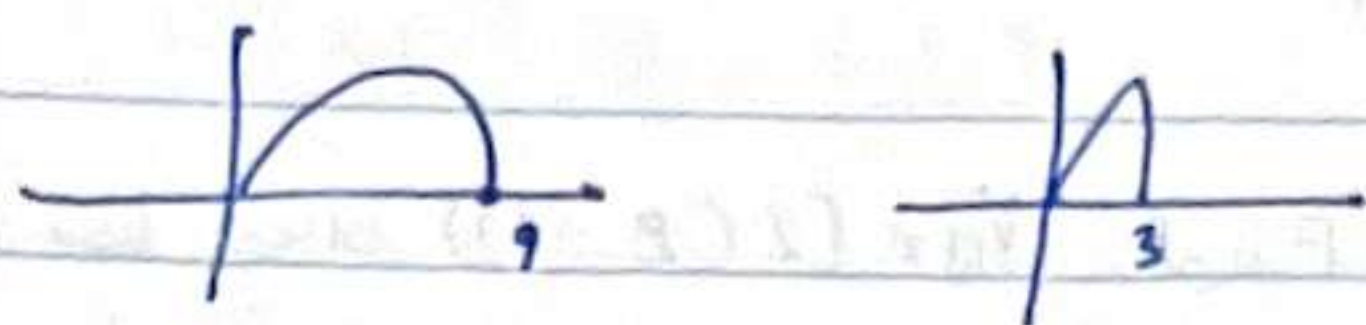
(a) Given that $\int_0^{2 \log_e(8)} f(x) dx = \frac{7}{2}$ and $f^{-1}(x) = \frac{1}{4}e^{\frac{x}{2}}$
Find shaded area.

数 10.2 Discrete random variables 离散随机变量

Q1 Unbiased die is tossed twice. Find the probability distribution for this discrete random variable.

let X be the number of sixes obtained

- know that the function is dilated by factor of $\frac{1}{3}$ from y-axis



Hence, Area is multiplied by $\frac{1}{3}$, $\int_0^3 f(3x) dx = \frac{5}{3}$

- 逻辑: shade area = $2 \log_e(8)$

找到了 黑色部分 = $\frac{7}{2}$ units²

hence, shaded area = Rectangle area - $\int_0^{2 \log_e(8)} f'(x) dx$
 $= 4 \log_e(8) - \frac{7}{2}$ units²

○

x	0	1	2
$\Pr(X=x)$	$\frac{25}{36}$	$\frac{10}{36} = \frac{5}{18}$	$\frac{1}{36}$

$\Pr(X=0)$: $\frac{5}{6} \times \frac{5}{6}$ 两次都在 1 到 5 之间 = $\frac{25}{36}$

$\Pr(X=1)$: $\frac{5}{6} \times \frac{1}{6} + \frac{1}{6} \times \frac{5}{6} = \frac{10}{36} = \frac{5}{18}$

第一次 1 到 5, 第二次是 6 第一次 6 第二次 1 到 5

$\Pr(X=2)$: $\frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$ 两次都是 6

中数 10.3 Measures of centre and spread

Q1 Given $\text{Var}(z) = 153.48$, Find $\text{Var}(2(z-1))$ and $\text{Var}(3-z)$

Q2 Given $E(X) = 1$ Find $E(5X+3)$

中数 10.4 Application

Q1. CAS Active, Given that $\Pr(V \cup W) = 0.7725$ and $\Pr(V \cap W) = 0.2275$

(a) Find $\Pr(V)$ and $\Pr(W)$, given that $\Pr(V) < \Pr(W)$ and events V and W are two independent events

(b) Draw a table and hence find $\Pr(V' \cap W')$

$$\begin{aligned} \text{Var}(2z-2) &= 2^2 \text{Var}(z) \\ &= 4 \times 153.48 = 613.91 \end{aligned}$$

$$\begin{aligned} \text{Var}(3-z) &= (-1)^2 \text{Var}(z) \\ &= 153.48 \end{aligned}$$

$$\begin{aligned} E(5x+3) &= 5E(x) + 3 \\ &= 5 + 3 = 8 \end{aligned}$$

10.4

$$\textcircled{a} \Pr(V \cup W) = \Pr(W) + \Pr(V) - \Pr(W \cap V)$$

$$0.7725 = \Pr(W) + \Pr(V) - 0.2275$$

$$\textcircled{1} 1 = \Pr(W) + \Pr(V)$$

$$\textcircled{2} 0.2275 = \Pr(W) \times \Pr(V)$$

根据条件: $\Pr(W) = 0.65$ $\Pr(V) = 0.35$

$\textcircled{b}$

	W	W'	
V	0.2275	0.1225	0.35
V'	0.4225	0.2275	0.65
	0.65	0.65 0.35	1

$$\Pr(V' \cap W') = 0.2275$$

11.3 中數 The binomial distribution

- a1. The chance for Ruby to score a goal is 0.75. What is the least number of shots that Ruby must attempt to ensure that the probability of her scoring at least 1 goal in a match is more than 0.95?
[CAS Active]

11.4 中數 Application

- a1. Edie is completing a test with 20 multiple choice of 20 questions. Each has 5 possible answers.
If Edie knows the answers to the first 4 questions but must randomly guess the answers to the other questions, find the probability that she correctly answers 10 or more questions. Give answer to 4 decimal places.
[CAS Active]

0

$$X \sim \text{Bi}(n, 0.75)$$

$$\Pr(X \geq 1) \geq 0.95$$

$$1 - \Pr(X = 0) \geq 0.95$$

$$1 - 0.25^n \geq 0.95$$

$$1 - 0.95 \geq 0.25^n$$

$$n \geq 2.18$$

3 shots would be required

原 $z \sim \text{Bi}(20, \frac{1}{5})$

Binomial CDF (10, 20, 20, $\frac{1}{5}$)

现

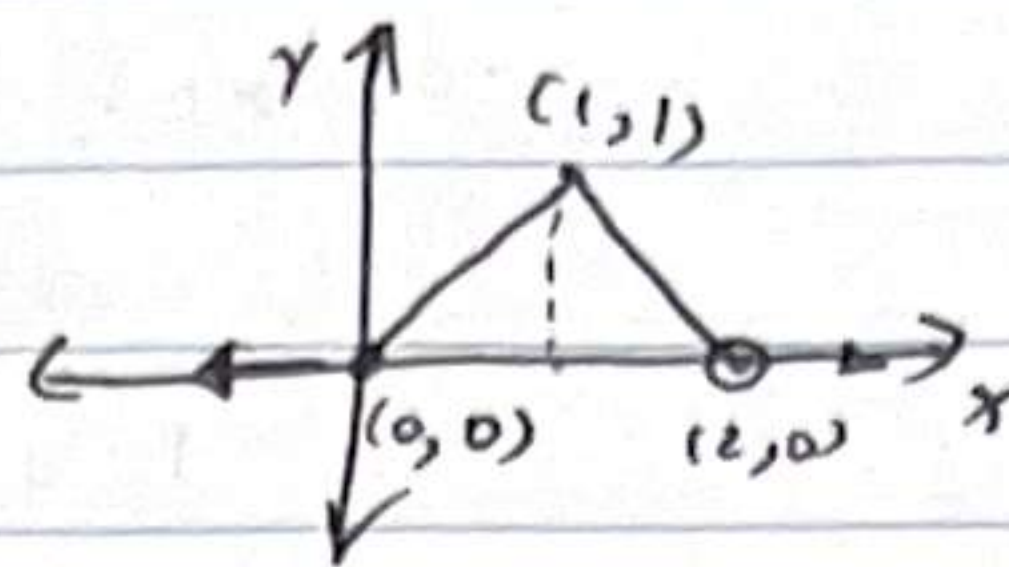
Binomial CDF (6, 16, 16, $\frac{1}{5}$)

z = correct times

$$\Pr(z \geq 6) \approx 0.0817$$

12.4 Continuous random variable.

$$f(x) = \begin{cases} x & 0 \leq x < 1 \\ 2-x & 1 \leq x < 2 \\ 0 & x < 0 \text{ or } x \geq 2 \end{cases}$$



Find m , the median of the $f(x)$

13.2 The normal distribution

~~Q1~~

Q1 Mean = 120 SD = 20, Find $\Pr(X < 80)$ ~~to 4 decimal~~ to 4 decimal
[CAS Active]

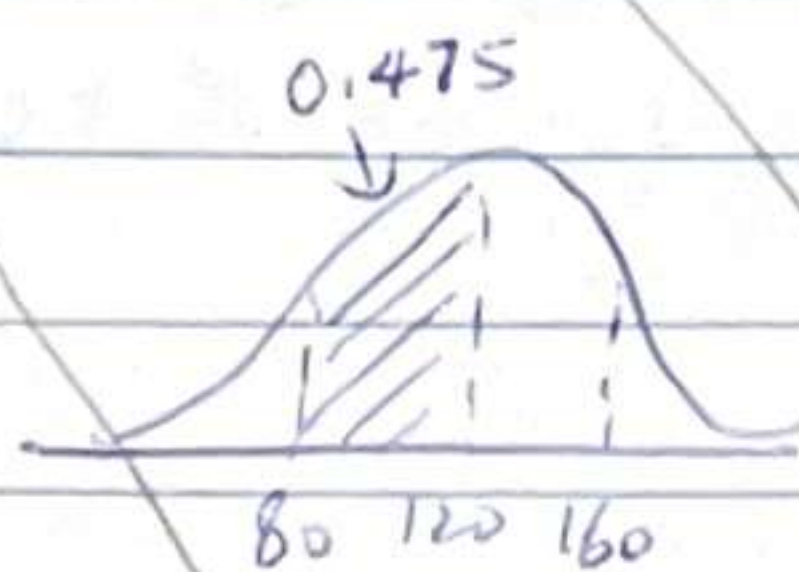
○ 从图中观察, 如果 Area 的一半发生在前半截 则用 $\int_1^m x dx = 0.5$
找 m

如果是在后半截, 则用 $\int_m^2 2-x dx = \frac{1}{2}$

在这题由于 median 发生在两个 graph 中间 (在 $x=1$) 所以两个都可以
哦! Ans: $m=1$ 刚好

○

$$\mu = 2\sigma = 80$$



$\text{Var}(X)$
↓

$$X \sim N(120, 20^2)$$

Interactive / Distribution / continuous / norm CDF

lower: $-\infty$

upper: 80

$\sigma = 20$

$\mu = 120$

$$\Pr(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 0.95$$

$$0.95 \div 2 = 0.475$$

$$\Pr(-\infty \leq X \leq 120) = 0.5$$

$$\therefore \Pr(X < 80) = 0.5 - 0.475 = 0.025$$

得 0.0228

13.3 Standard normal distribution $z = \frac{x - \mu}{\sigma}$

↑↑↑↑

Q1 $X \sim N(5, 3^2)$, find b such that $\Pr(X > 7) = \Pr(Z < b)$

↓↑↑↑↑

Q2 $X \sim N(38, 2.4^2)$, if $\overset{\text{score}}{\uparrow} X = 39.5$, determine if it is in the top 15%
[CAS active]

↑↑↑↑

Q3. Jing $X \sim N(72, 9^2)$ score 85

Nani $Y \sim N(15, 4^2)$ score 18, which one has better performance?

we z

○

$$z = \frac{x - \mu}{\sigma}$$

$$= \frac{7.5}{3}$$

$$Pr(X > 7) = Pr(z > \frac{2}{3})$$

$$= Pr(z < -\frac{2}{3})$$

$$= \frac{2}{3}$$

2222身, $b = -\frac{2}{3}$

○

$$Pr(X \geq 39.5) \approx 0.2659$$

norm CDF (39.5, ∞ , 2.4, 38)

它在 top 27% 不在 top 15%

○

$$\text{Jing: } z = \frac{85 - 72}{9}$$

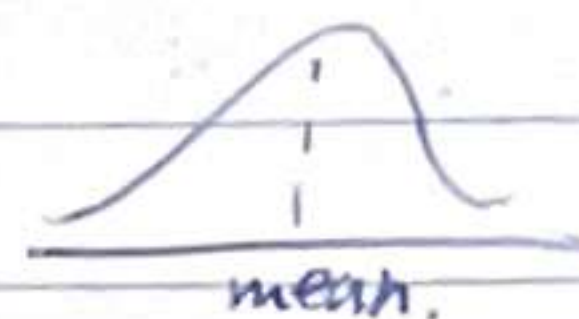
$$= 1.4$$

→ Mean 往右走 1.4

$$\text{Rani: } z = \frac{18 - 15}{4}$$

$$= 0.75$$

→ Mean 往右走 0.75



13.4 The inverse normal distribution [CAS Active]

Q ↑↑↑↑↑

Q1 $X \sim N(37.5, 8.62^2)$, Find a to 2 decimal

[CAS Active]

$$\Pr(37.5 - a < X < 37.5 + a) = 0.88$$

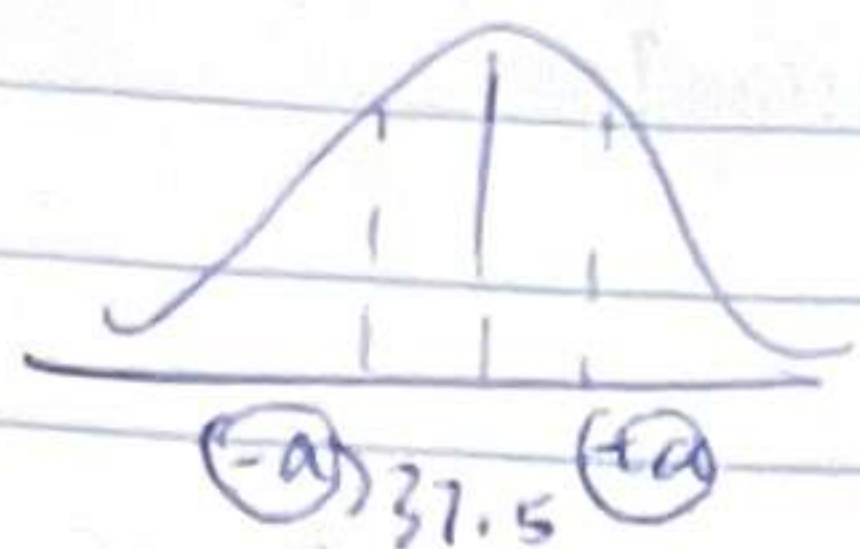
↑↑↑↑↑

Q2. $X \sim N(\mu, 4.45^2)$. If 11th quantile is 32.142. find μ to 1 decimal

↓
必须用 $z = \frac{x - \mu}{\sigma}$

[CAL Active]

○



$$0.88 \div 2 = 0.44$$

Interactive/Distribution/Inverse

$$0.5 - 0.44 = 0.06$$

↑

Find 37.5 - a: InvNormCDF("L", 0.06, 8.62, 37.5)

$$37.5 - a = 24.10$$

$$a \approx 13.40$$

○

$$\Pr(X < 32.142) = 0.11$$

$$\Pr(Z < z) = 0.11$$

Use standard normal distribution # z: Interactive/Distribution/Inverse/

InvNormCDF

InvNormCDF("L", 0.11, 1, 0)

z of σ and mean

$$z \approx -1.2265$$

$$z = \frac{x - \mu}{\sigma}$$

$$-1.2265 = \frac{32.142 - \mu}{4.45}$$

$$\mu \approx 37.6$$

13.5 中数 Mixed probability application problems [(AS Active)]

↑↑↑↑↓

Q1 机器零件合格检测, $\text{Weight} \sim N(0, 0.76^2)$

The parts are considered unacceptable if the error is more than 1.5 grams

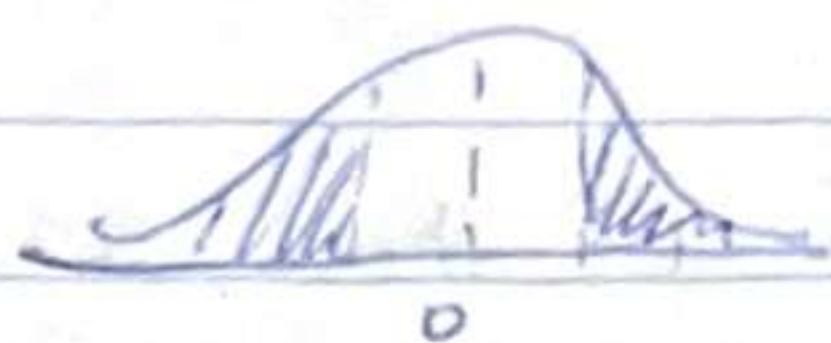
Find Proportion of parts which are unacceptable, to 4 decimals

↑↑↓↑↑

Q2 A Large bottle has Mean 1.015L and SD of 0.009L.

Find the probability that a box of 12 randomly chosen large bottles contain at least 4 bottles whose contents are each less than 1 Litre, to 4 decimal

$$\circ \quad \Pr(\text{unacceptable}) = \Pr(W < 1.5) \cup \Pr(W > 1.5)$$



$$\text{NormCDF}(1.5, \infty, 0.76, 0)$$

$$\approx 0.0242$$

$$\text{Symmetry property: } 0.0242 \times 2 = 0.0484$$

$$\text{Ans: } 0.0484$$

$$\circ \quad \text{先认识到这是一个 binomial distribution, } L \sim B(n, p) \quad \begin{matrix} n, p \\ \downarrow \\ \text{未知} \end{matrix}$$

$$\text{用 normal distribution 找 } p: \Pr(L < 1) \approx 0.0478$$

$$\downarrow \text{normCDF}(-\infty, 1, 0.009, 1.015)$$

$$\text{然后用 binomial 找答案: binomial CDF}(\underbrace{4}_{\text{大于或等于4个}}, \underbrace{12}_{n}, 0.0478)$$

$$\approx 0.0019$$

14.3 中數 The distribution of $\hat{p}$

↑↑↑↑

Q1 There are thousands of pens. Of these, 25% are blue.

A sample of 24 pens are selected. Find $\Pr(\hat{p} \geq \frac{5}{24})$, to 4 decimal places

↑↑↑↑↑

Q2 Four people are selected for a sample.

If $\Pr(X=4) = \frac{1}{625}$, find the $\Pr(\hat{p} < 0.5)$ to 4 decimal places.

○ let X be the number of blue pens in the sample

$$\hat{p} = \frac{X}{n}$$

$$\frac{5}{24} = \frac{X}{24}, X=5 \text{ if } \hat{p} = \frac{5}{24}$$

∴ Find $\Pr(X \geq 5)$

$$X \sim \text{Bi}(24, 0.25)$$

$$(n, p)$$

$$\Pr(X \geq 5) \approx 0.7534$$

○ ① $\Pr(X=4) = {}^4C_4 \cdot (1-p)^0 \cdot p^4 = \frac{1}{625}$

$$p = \frac{1}{25} = 0.2$$

② Convert: $\Pr(\hat{p} < 0.5)$

$$= \Pr\left(\frac{X}{4} < 0.5\right)$$

$$= \Pr(X < 2)$$

$$= \Pr(X \leq 1)$$

$$X \sim \text{Bi}(4, 0.2)$$

$$\text{binomialCDF}(0, 1, 4, 0.2) \approx 0.8192$$

$$\begin{array}{ccccccc} \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ \text{lower} & \text{upper} & n & p \end{array}$$